

TASK 1

```
jella_pragathi@LAPTOP-RQQT555:~$ gcc --version
git --version
make --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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git version 2.53.0
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
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License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
jella_pragathi@LAPTOP-RQQT555:~$ mkdir -p ~/ossp_project
jella_pragathi@LAPTOP-RQQT555:~$ cd ~/ossp_project
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ mkdir src include build
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ touch src/main.c
touch Makefile
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ ls
Makefile  build  include  src
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ tree
.
├── Makefile
├── build
├── include
└── src
    └── main.c

4 directories, 2 files
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ nano src/main.c
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ cat src/main.c
#include <stdio.h>
int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
```

```
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ nano Makefile
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ cat Makefile
CC = gcc
CFLAGS = -Wall -Wextra -g
TARGET = build/main
SRC = src/main.c
all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)
clean:
    rm -f $(TARGET)
run: all
    ./${TARGET}
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ make
gcc -Wall -Wextra -g src/main.c -o build/main
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ ls build
main
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ ./build/main
OSSP Project Started Successfully!
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ |
```

TASK 2

```

jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/process_demo.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ gcc -Wall -Wextra -g src/process_demo.c -o build/process_demo
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/process_demo
Parent Process Started
Parent PID: 12917

Parent waiting for child...

Child Process Created
Child PID: 12918
Parent PID: 12917
Child is executing ls command...
total 16
-rw-r--r-- 1 jella_pragathi jella_pragathi 160 Aug 17 05:32 Makefile
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:39 build
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:29 include
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:39 src
Child process completed.
Parent process exiting.
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ pstree -p
systemd(1)
├──agetty(252)
├──chronyd-starter(108)──chronyd(231)──chronyd(258)
├──cron(109)
├──dbus-daemon(110)
├──init-systemd(Ub(2))
│   ├──SessionLeader(12604)──Relay(12612)(12605+
│   │   ├──init(7)──{init}(8)
│   │   ├──login(338)──bash(432)
│   │   └──init-systemd(Ub)(9)
├──networkd-dispat(168)
├──polkitd(709)
│   ├──{polkitd}(711)
│   ├──{polkitd}(712)
│   └──{polkitd}(713)
├──rsyslogd(3206)
│   ├──{rsyslogd}(3207)
│   ├──{rsyslogd}(3208)
│   └──{rsyslogd}(3209)
├──systemd(399)
│   ├──(sd-pam)(403)
│   └──dbus-daemon(4018)
├──systemd-journal(1501)
├──systemd-logind(179)
├──systemd-resolve(1628)
├──systemd-udev(1564)
└──unattended-upgr(268)──{unattended-upgr}(310)
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ |

```

TASK 3

```

jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ~/ossp_project
-bash: /home/jella_pragathi/ossp_project: Is a directory
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/myshell.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ gcc -Wall -Wextra -g src/myshell.c -o build/myshell
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/myshell
Mini Shell Started
Type 'exit' to quit.

myshell> hello
You entered: hello
myshell> exit
Exiting shell...
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ |

```

TASK 4

```
Exiting shell...
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/keyboard_shell.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ gcc -Wall -Wextra -g src/keyboard_shell.c -o build/keyboard_shell
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/keyboard_shell
Keyboard Input Shell
Type commands and press Enter.
Use Backspace to edit.
Type 'exit' to quit.

myshell> hello
Command entered: hello
myshell> exit
Exiting shell...
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ tree
.
├── Makefile
├── build
│   ├── keyboard_shell
│   ├── main
│   ├── myshell
│   └── process_demo
├── include
└── src
    ├── keyboard_shell.c
    ├── main.c
    ├── myshell.c
    └── process_demo.c

4 directories, 9 files
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ |
```